

From Optimal Control to Diffusion Policy

A Trajectory-Optimization $\rightarrow$ TVLQR $\rightarrow$ Behavioural-Cloning Pipeline for
Double-Pendulum Swing-Up

Group Project TN2 — Technical Documentation

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1 Introduction

The goal of this project is to compare the performance of 2 different learning- based control methods. The first method is a behavioural cloning approach, where we use a dataset of expert demonstrations to train a policy that mimics the expert’s actions. The second method is a diffusion policy approach, where we use a diffusion model to learn a policy that can generate actions based on the current state of the system. We will evaluate the performance of both methods in the **double pendulum swingup task** and the **stacking problem**.

1.1 Behavioural Cloning

1.2 Diffusion

In order to understand the diffusion policy approach, we first need to understand the concept of diffusion models. Diffusion models are a class of generative models that learn to generate structured data (in our case, action sequences) by iteratively denoising a sample of noise.

Parameter Setup

For the diffusion model, we first need to define the time parameters of the noising and denoising process.

- **Time Horizon** T is the number of time steps in the diffusion process.
- **β Schedule** β_t is a set of hyperparameters that control how quickly the noise is added to the data during the forward diffusion process. It is a monotonic increasing sequence of values that determine the variance of the noise
- **Noise Schedule** $\alpha_t = 1 - \beta_t$ is a set of hyperparameters that control how quickly the noise is removed.
- **Cumulative Noise Schedule** $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$
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